1 A changing e.m.f. in a coil can induce an e.m.f. in another coil.

For Examiner's Use

Fig. 1.1 shows a coil (coil X), which is wound on a cardboard tube.

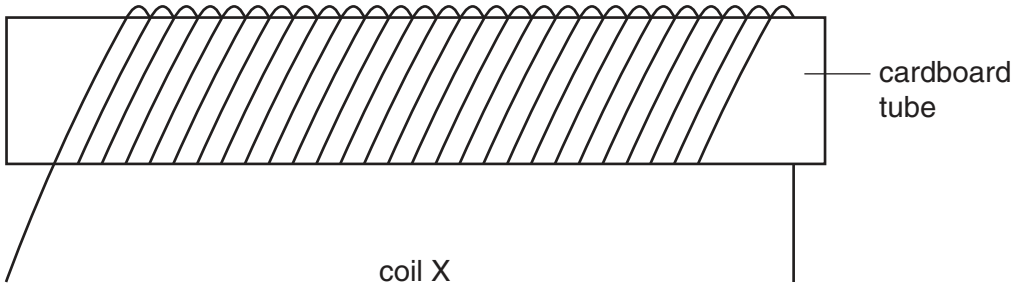

Fig. 1.1

Coil X has cross-sectional area *A*.

A student winds another coil (coil Y) tightly around coil X. The student wishes to investigate how the e.m.f. *V* in coil Y depends on *A*.

It is suggested that *V* is directly proportional to *A*.

Design a laboratory experiment to investigate the suggested relationship. You should draw, on page 3, a diagram showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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